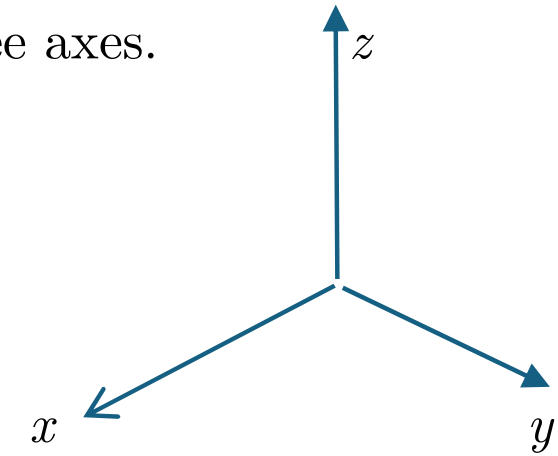
Triple integrals

F18CE Week 10

Sketching 3D graph

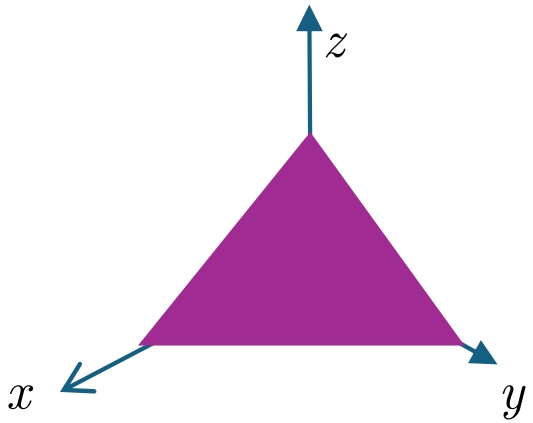
1. Plotting the three axes.



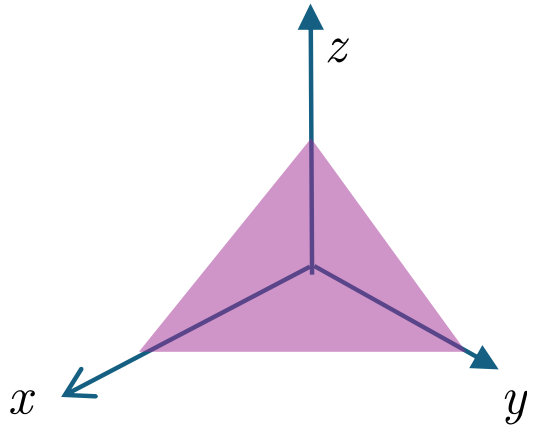
View from top you see the xy-plane (the floor).

2. $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1\,,\ a,b,c\neq 0$ is a plane. We draw a triangle plane with edges on the axes.

See directly:

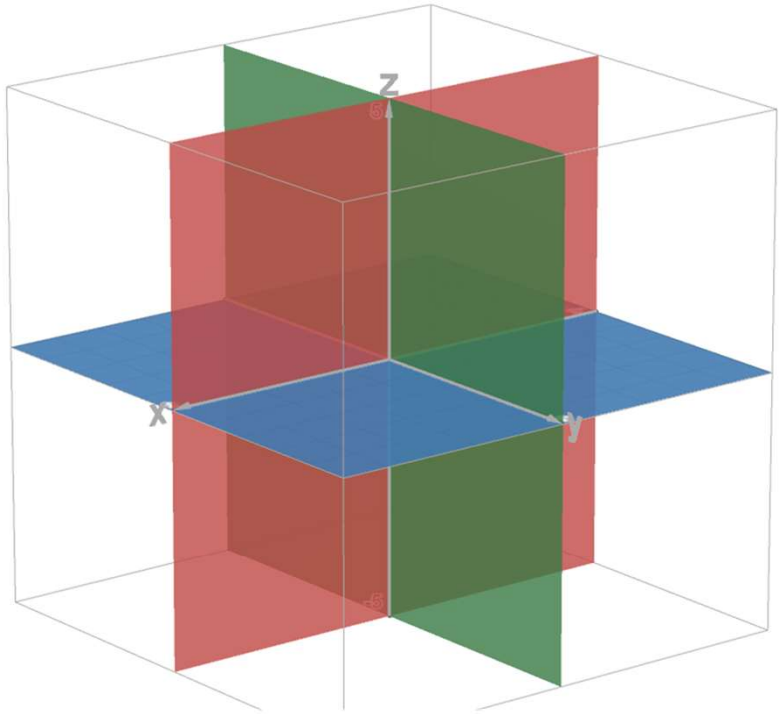


See through:



3. xy-plane (the floor) is when $z = 0$; same goes for the other two (the side walls).

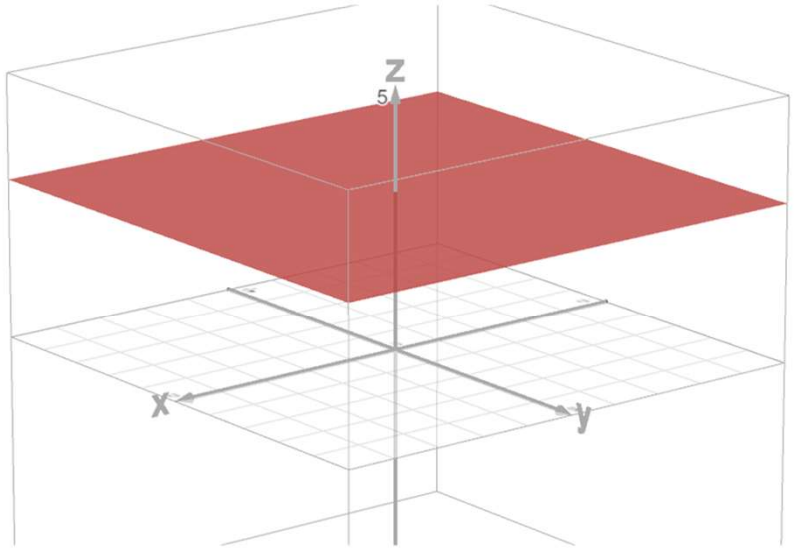
In general, $z = k$ is a plane parallel to the xy-plane; same goes for the other two.



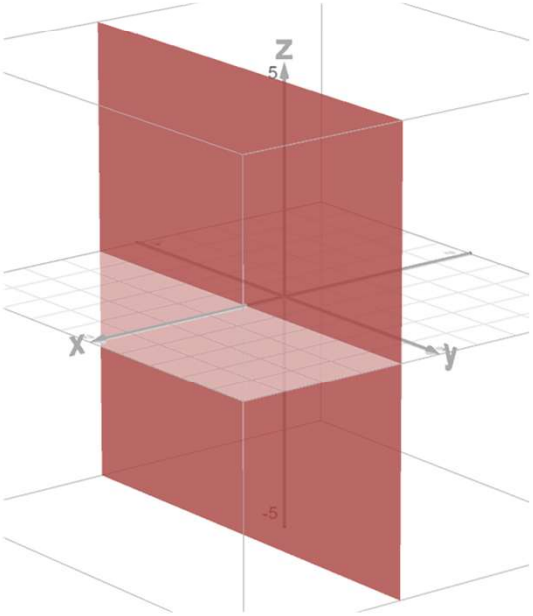
Red is the xz-plane, i.e., $y = 0$

Green is the yz-plane, i.e., $x = 0$

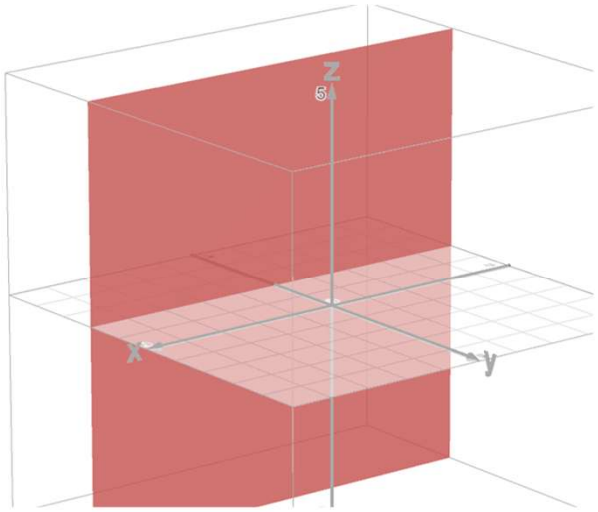
Blue is the xy-plane, i.e., $z = 0$



$z = 3$

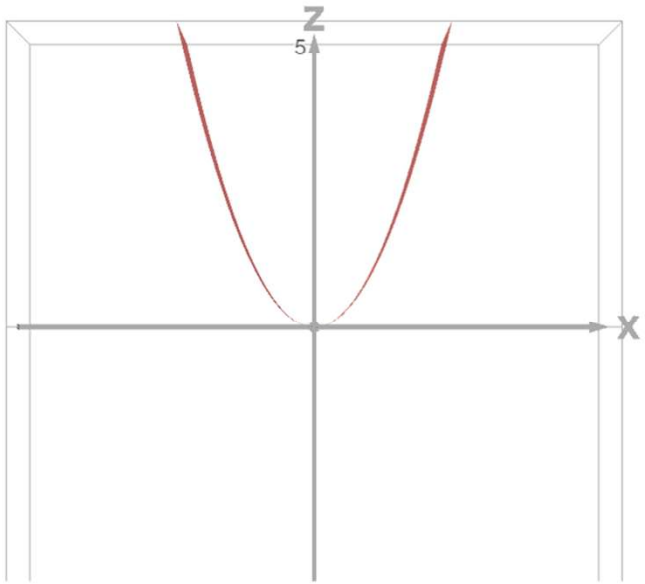


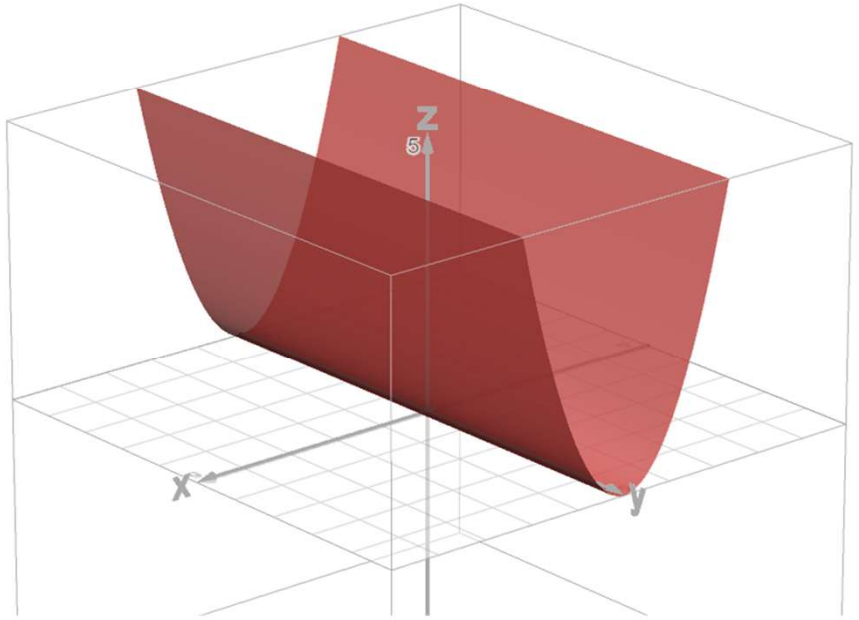
$x = 1$



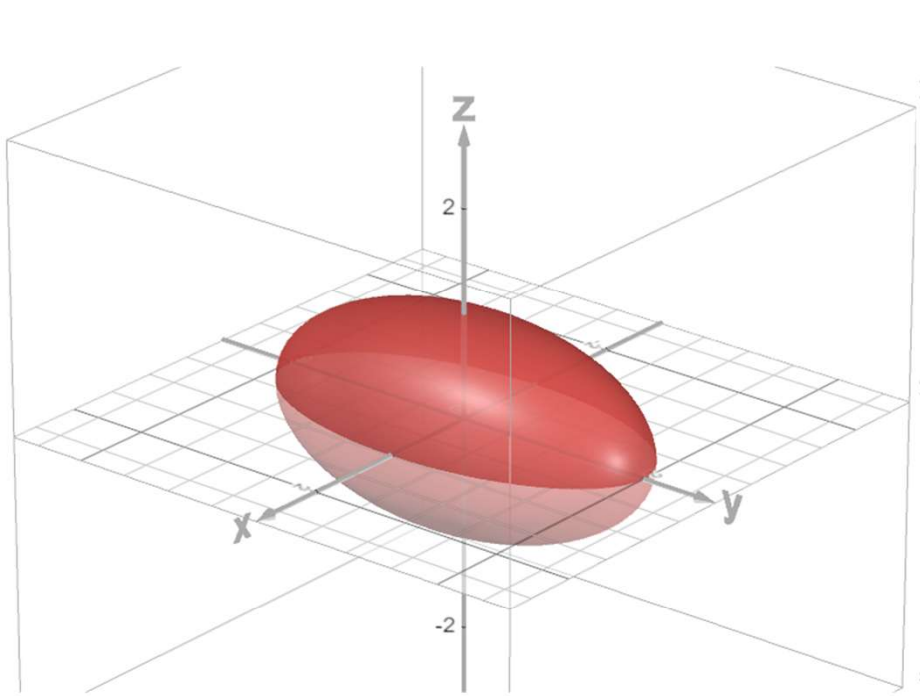
$y = -1$

4. If $y = f(x)$; then the 3D graph is projected from xy-plane; same goes for the other two. For example, $z = x^2$ is the following (projected from xz-plane):

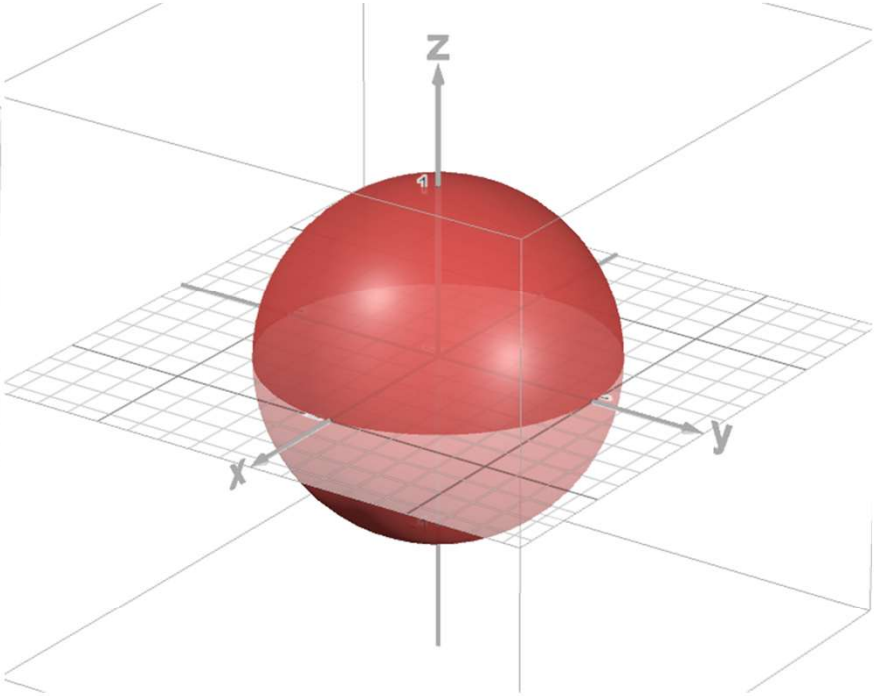




5. $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is an ellipsoid. A special case of this is a sphere (ball) when $a = b = c$.

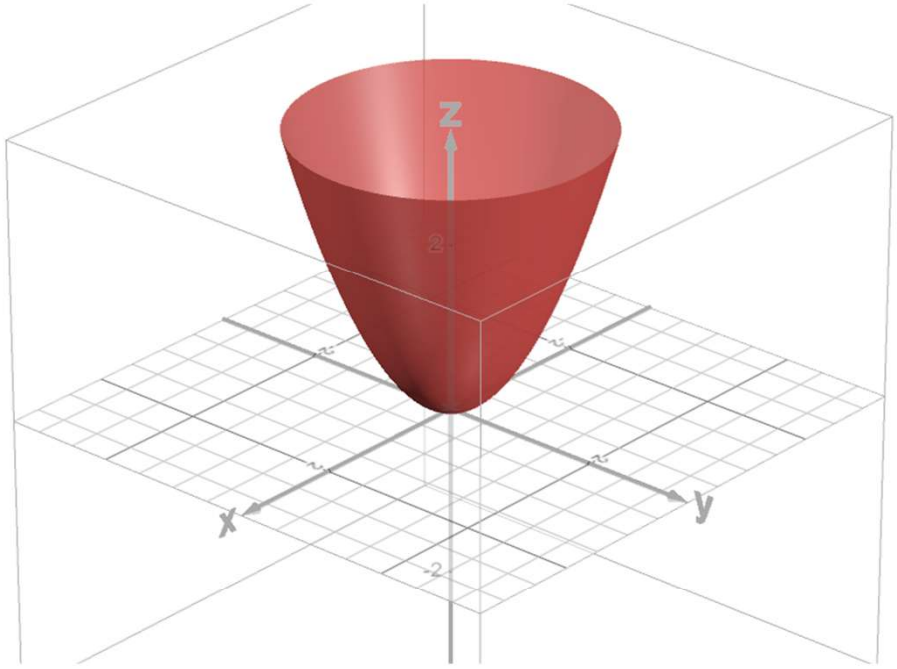


Ellipsoid

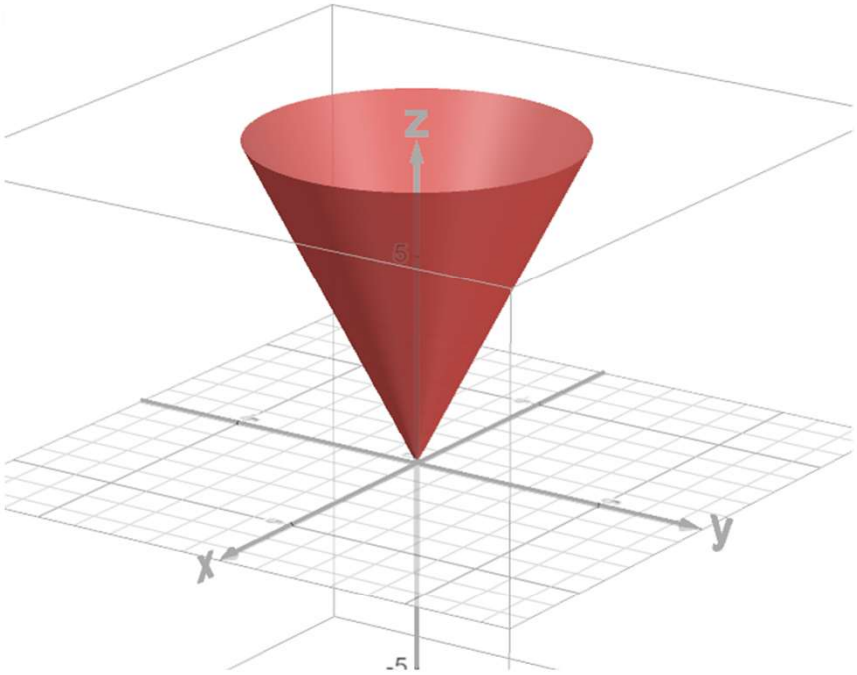


Sphere (ball)

6. $z = x^2 + y^2$ is paraboloid. $z = \sqrt{x^2 + y^2}$ is a cone with the edge (sharp point) facing downward ($z = -\sqrt{x^2 + y^2}$ has the edge facing upward).



Paraboloid $z = x^2 + y^2$



Cone $z = \sqrt{x^2 + y^2}$

Exercise

7. All the above can be modified. Try to sketch

(a) $z = -(x^2 + y^2) + 1$ (b) $z = 10 - \sqrt{x^2 + y^2}$

Notations *dA*, *dV*

Let A be denoted as area, we sometimes write $dydx$

(multiplication of changes in x and in y is equivalent to changes in A)

as dA, i.e., $\iint_D f(x,y)dydx = \iint_D f(x,y)dA\,.$

Similarly, let V be denoted as volume, and write $dzdydx$ as dV, i.e.,

$\iiint_D f(x,y,z)dzdydx = \iiint_D f(x,y,z)dV\,.$

Definition 9.1: Bounded region in three dimensions

A 3-dimensional region $D \subset \mathbb{R}^3$ is said to be bounded if $\exists R > 0$ such that $D \subset B_R = \left\{ \left(x, y, z \right) : x^2 + y^2 + z^2 < R^2 \right\}$.

In other words, D is said to be bounded if it fits inside an open ball B_R of sufficiently large radius R.

Interchanging of variables in triple integrals

$$\iiint_D f(x,y,z)\,dV = \underbrace{\int_{x=x_0}^{x=x_1} \underbrace{\int_{y=y_0}^{y=y_1} \overbrace{\int_{z=z_0}^{z=z_1} f(x,y,z)dz}^{\text{Function of } x,y}\, dy}_{\text{Function of only } x}\, dx}\,.$$

In this case we integrate with respect to z first, followed by y and then x finally. We have five more other possible ways to do so, e.g. $dxdydz$, $dxdzdy$, etc.

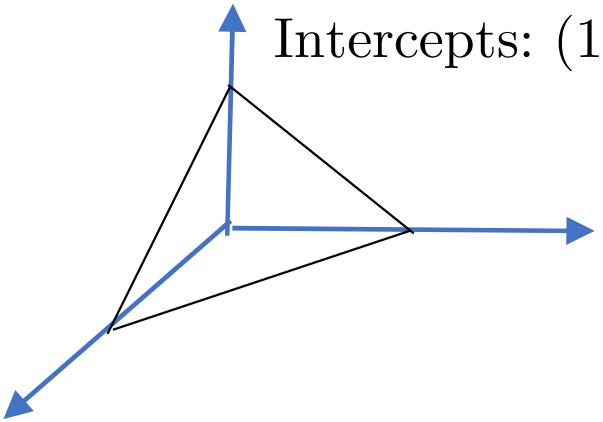
Example 9.3

Let D be the region in the first octant bounded by the coordinate planes and the plane $x + y + z = 1$. Find $\iiint_D x \, dV$.

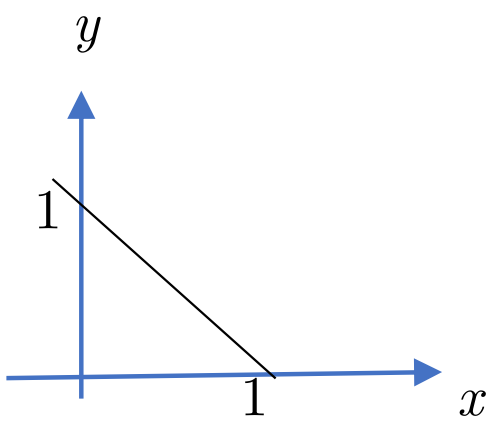
Solution.

Sketch the plane $x + y + z = 1$ and also consider the xy-plane. We have:

$$0 \le z \le 1 - x - y, \ 0 \le y \le 1 - x \ \text{ in } \ 0 \le x \le 1$$



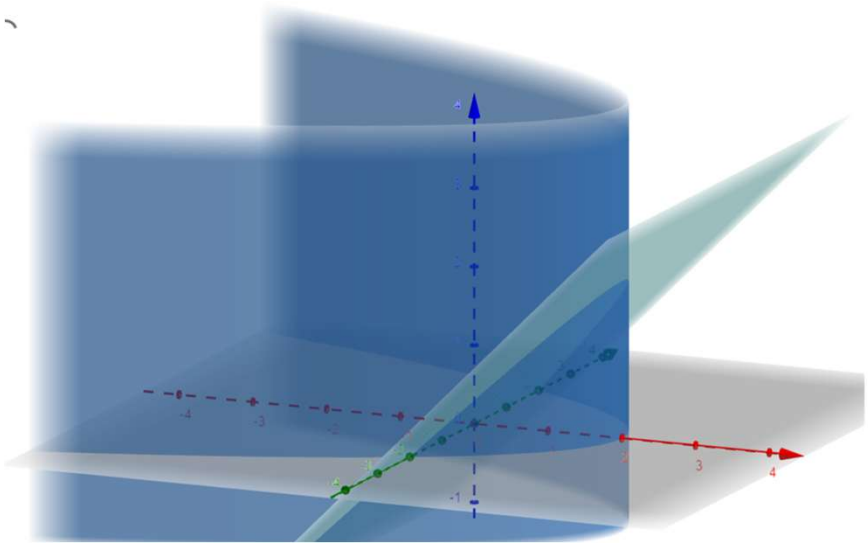
Intercepts: (1, 0, 0), (0, 1, 0), (0, 0, 1).

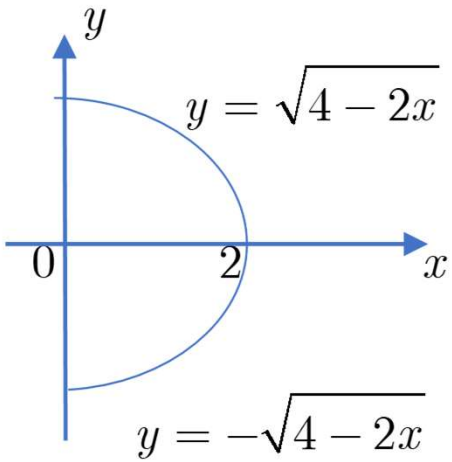


$$
\begin{aligned}
\iiint_D x\,dV &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} x\,dz\,dy\,dx \\
&= \int_0^1 \int_0^{1-x} xz\Big|_0^{1-x-y} dydx \\
&= \int_0^1 \int_0^{1-x} x(1-x-y)\,dydx \\
&= \int_0^1 xy - x^2y - \tfrac{1}{2}xy^2\Big|_0^{1-x} dx \\
&= \int_0^1 \Big(x(1-x) - x^2(1-x) - \tfrac{1}{2}x(1-x)^2\Big)\,dx \\
&= \Big[\tfrac{1}{4}x^2 - \tfrac{1}{3}x^3 + \tfrac{1}{8}x^4\Big]_0^1 \\
&= \tfrac{1}{24}
\end{aligned}
$$

Example 9.4

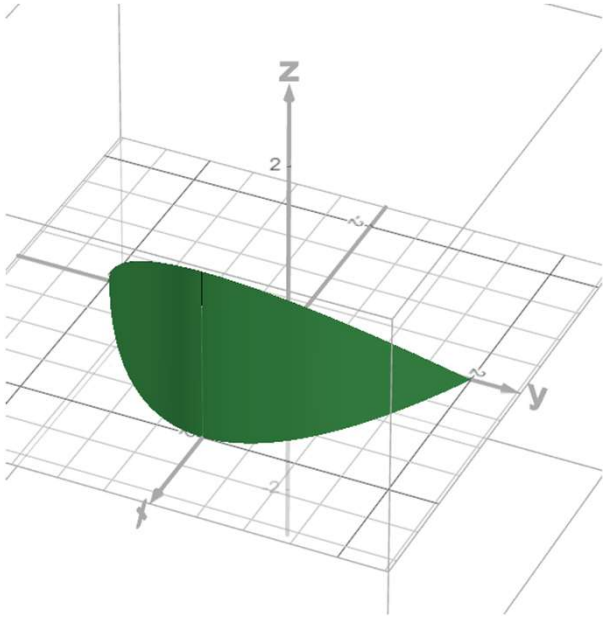
Express $\iiint_D f(x,y,z)\,dV$ as $\int_{x_0}^{x_1}\int_{y_0}^{y_1}\int_{z_0}^{z_1} f(x,y,z)\,dz\,dy\,dx$ if D is the region bounded by $z=0$, $z=x$ (planes) and $y^2=4-2x$ (half-cylinder) and with $x\geq 0$.

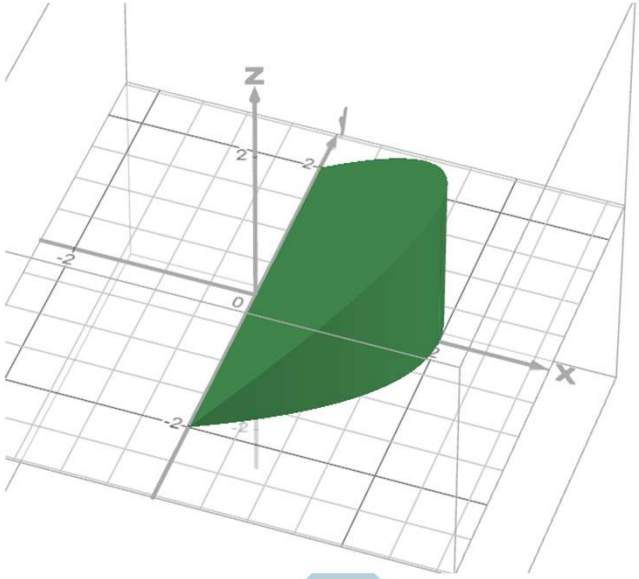




We have $0 \le z \le x$, $-\sqrt{4-2x} \le y \le \sqrt{4-2x}$ in $0 \le x \le 2$. Then

$$\iiint_D f(x,y,z)\,dV = \int_0^2 \int_{-\sqrt{4-2x}}^{\sqrt{4-2x}} \int_0^x f(x,y,z)\,dz\,dy\,dx = 2\int_0^2 \int_0^{\sqrt{4-2x}} \int_0^x f(x,y,z)\,dz\,dy\,dx$$





Apply triple integral to find volume

Volume generated by the solid S is given by

$$\boxed{\iiint_S 1 dz dy dx}$$

Example 9.5

Suppose $f = 1$ for Example 9.4. Then we can compute the volume of the solid generated.

$$\begin{aligned}
\iiint_D 1 dV &= 2\int_0^2 \int_0^{\sqrt{4-2x}} \int_0^x 1 dz dy dx = 2\int_0^2 \int_0^{\sqrt{4-2x}} \Big[z\Big]_0^x dy dx \\
&= 2\int_0^2 \Big[xy\Big]_0^{\sqrt{4-2x}} dx = 2\int_0^2 x\sqrt{4-2x} dx \\
&= \tfrac{2}{3}\Big[-x(4-2x)^{\frac{3}{2}}\Big]_0^2 + \tfrac{2}{3}\int_0^2 (4-2x)^{\frac{3}{2}} dx \\
&= 0 - \tfrac{2}{3}\left(\tfrac{2}{5}\right)\left(\tfrac{1}{2}\right)(4-2x)^{\frac{5}{2}}\Big|_0^2 = \tfrac{64}{15}
\end{aligned}$$

Alternative (using double integral):

$$V = 2\int_0^2 \int_0^{\sqrt{4-2x}} z \; dydx = 2\int_0^2 \int_0^{\sqrt{4-2x}} x \; dydx$$

SUMMARY

	Finding area	Finding volume
Single integral, $f : \mathbb{R} \to \mathbb{R}$ $(y = f(x))$	$\int_D f(x)\,dx$	$\int_D \pi\left(f(x)\right)^2 dx$
Double integral, $f : \mathbb{R}^2 \to \mathbb{R}$ $(z = f(x,y))$	$\iint_D 1\,dxdy$	$\iint_D f(x,y)\,dxdy$ *
Triple integral, $f : \mathbb{R}^3 \to \mathbb{R}$ $(w = f(x,y,z))$	-	$\iiint_D 1\,dxdydz$

* Solid generated from $z = 0$ to $z = f\,(x,\ y)$.

General substitution in triple integrals

Let $f(x,y,z) : \mathbb{R}^3 \to \mathbb{R}$ be a function and write $x = x(u,v,w)$, $y = y(u,v,w)$, and $z = z(u,v,w)$. Then

$$\iiint_V f(x,y,z)\,dxdydz = \iiint_{V'} f(x(u,v,w), y(u,v,w), z(u,v,w))\left|J\right| dudvdw,$$

where $\left|J\right|$ is the absolute value of the Jacobian

$$J = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix}.$$

Remark: If $f(x,y,z) = f(x)g(y)h(z)$, i.e., function f can be written independently as product of functions in x, y and z; and $x_0, x_1, y_0, y_1, z_0, z_1$ are constants. Then

$$\int_{z_0}^{z_1}\int_{y_0}^{y_1}\int_{x_0}^{x_1} f(x,y,z)\,dxdydz = \int_{x_0}^{x_1} f(x)\,dx \int_{y_0}^{y_1} g(y)\,dy \int_{z_0}^{z_1} h(z)\,dz\,.$$

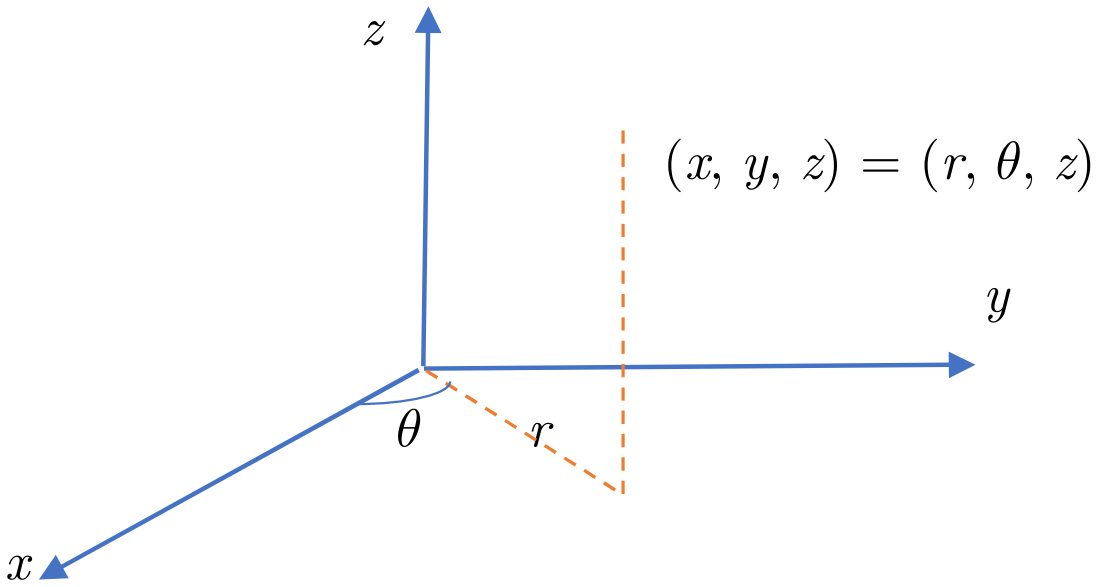
Special Substitution: Cylindrical polar coordinates

Let $x = r\cos\theta,\ y = r\sin\theta,\ z = z\,(\text{unchanged}).$ Then

$$J = \det\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial z} \end{pmatrix} = \det\begin{pmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = r\,.$$

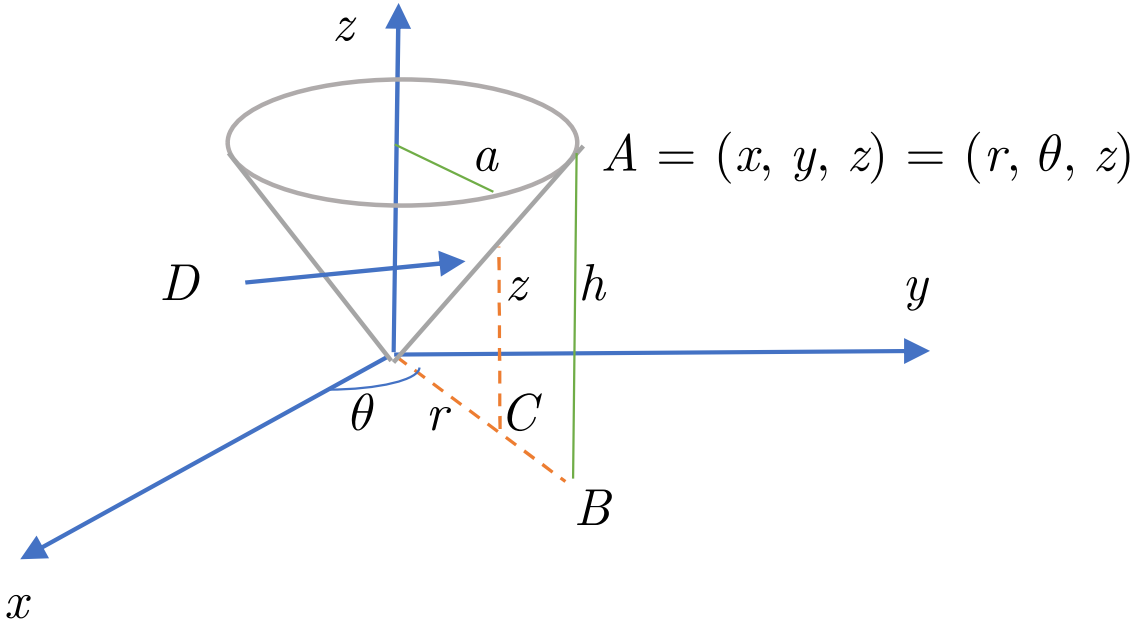
Therefore,

$$\iiint_V f(x,y,z)\,dxdydz = \iiint_{V'} f(x(r,\theta),y(r,\theta),z)\,rdrd\theta dz\,.$$

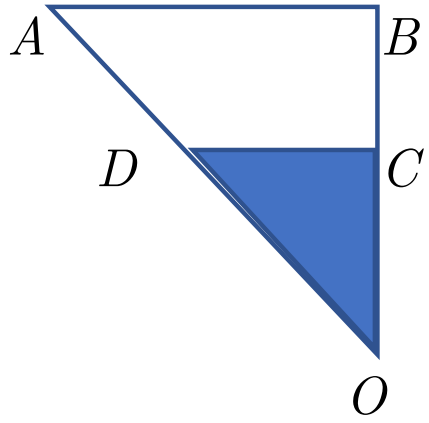

z
$(x,\ y,\ z) = (r,\ \theta,\ z)$
y
θ
r
x

Example 9.6: Volume of a cone with radius *a* and height *h*

Let $x = r\cos\theta$, $y = r\sin\theta$, $z = z$ (unchanged). Then $dxdydz = rdrd\theta dz$. A, B, C, D are points on the 3D plane and O as the origin.



Consider the similar triangles ABO and DCO, so $\dfrac{AB}{OB} = \dfrac{DC}{OC} \Rightarrow \dfrac{h}{a} = \dfrac{z}{r} \Rightarrow r = \dfrac{az}{h}$.



Now we see that $0 \le r \le \frac{az}{h}$, $0 \le \theta \le 2\pi$, $0 \le z \le h$. Hence, the volume of the cone is

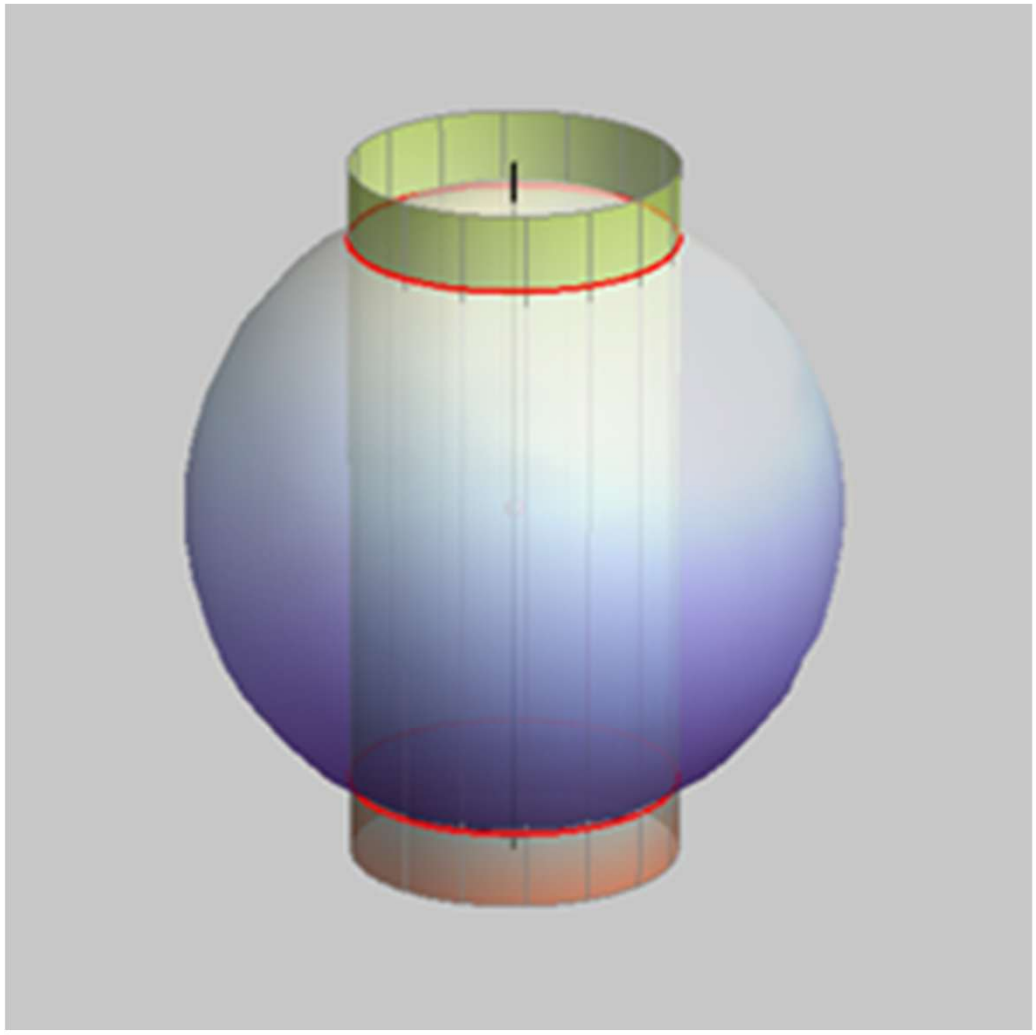
$$\iiint_D 1\,dxdydz = \iiint_{D'} 1 \cdot r dr dz d\theta = \int_0^{2\pi}\int_0^{h}\int_0^{az/h} r dr dz d\theta = \int_0^{2\pi}\int_0^{h} \tfrac{1}{2}\left(\tfrac{az}{h}\right)^2 dz\, d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{h} \tfrac{1}{2}\left(\tfrac{az}{h}\right)^2 dz = 2\pi \times \tfrac{1}{6}\left(\tfrac{a}{h}\right)^2 h^3 = \tfrac{1}{3}a^2 h\pi$$

Example 9.7

Find the volume of the region in the intersection of the solid cylinder $x^2 + y^2 \leq 1$ and the solid sphere described by $x^2 + y^2 + z^2 \leq 4$.

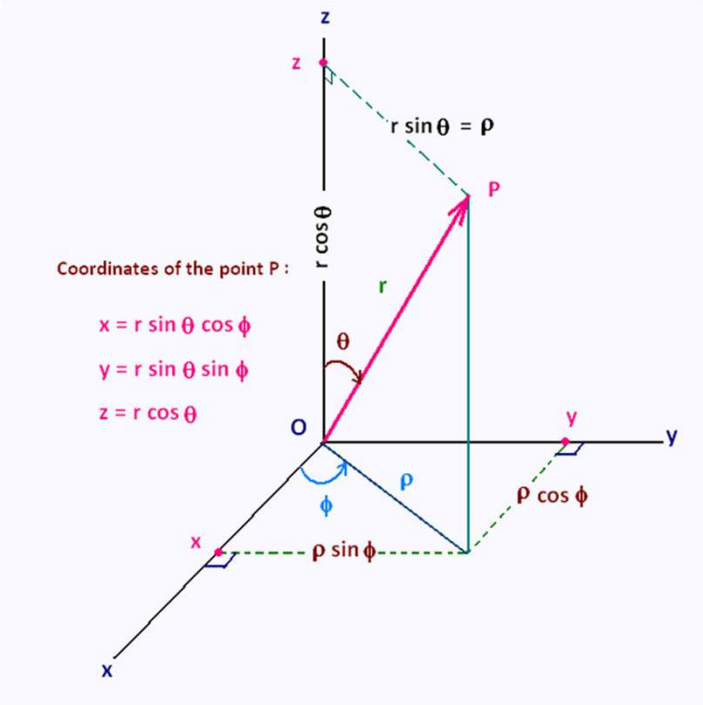
Let $x = r\cos\theta$, $y = r\sin\theta$, $z = z$ (unchanged). Then $dxdydz = rdrd\theta dz$ and $x^2 + y^2 \leq 1$ implies that $(0 \leq) r \leq 1$ and $x^2 + y^2 + z^2 \leq 4$ implies that $z^2 \leq 4 - r^2$. We will only consider one of the first octant $\left(0 \leq \theta \leq \frac{\pi}{2}\right)$ and multiple the answer by 8.



$$V = \iiint_D 1\,dxdydz = \iiint_{D'} 1 \cdot r dz dr d\theta = 8\int_0^{\pi/2}\int_0^{1}\int_0^{\sqrt{4-r^2}} r dz dr d\theta$$

$$= 8\int_0^{\pi/2}\int_0^{1} r\sqrt{4-r^2}\,dr\,d\theta = 8\int_0^{\pi/2} d\theta \times \frac{1}{3}\left[-\sqrt{(4-r^2)^3}\right]_0^1 = \frac{4}{3}\pi(8-3\sqrt{3})$$

Special Substitution: Spherical polar coordinates



Substitute $x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta$ in $0 \le \theta \le \pi$ and $0 \le \phi \le 2\pi$

Remark: To use this substitution, the solid must have a ball shape or part of it.

Substitute $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$ in $0 \le \theta \le \pi$ and $0 \le \phi \le 2\pi$,

$$\text{with } J = \det\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{pmatrix} = \det\begin{pmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix} = r^2\sin\theta\,.$$

Remark: Some authors interchange θ and ϕ, which is, in my opinion, more suitable, because the (old) idea of θ is remained and we introduce the new idea of ϕ. However, here we follow the arrangement in the lecture notes.

From Linear Algebra, $\det\begin{pmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix}$ is equal to

$$r^2\cos\theta\left(\cos\theta\sin\theta\cos^2\phi+\sin\theta\cos\theta\sin^2\phi\right)+r^2\sin\theta\left(\sin^2\theta\cos^2\phi+\sin^2\theta\sin^2\phi\right)$$
$$=r^2\cos^2\theta\sin\theta+r^2\sin^3\theta=r^2\sin\theta(\cos^2\theta+\sin^2\theta)=r^2\sin\theta$$

In view of Theorem 8.14, we now have

$$\iiint_S f\Big(x,y,z\Big)\,dV = \iiint_{S'} f\Big(x(r,\theta,\phi),y(r,\theta,\phi),z(r,\theta,\phi)\Big)\,r^2\,\Big|\sin\theta\Big|\,dr\,d\theta\,d\phi$$

Example 9.8

Evaluate $\iiint_{V'} x\,dV$, where V' is the part (one-eighth) of the solid ball $x^2 + y^2 + z^2 \leq a^2$ lying in the first octant.

Note that this question is not finding the volume of the one-eighth of the ball but just pure computation of triple integral of x with respect to x, y and z, by substitution of r, θ, ϕ, i.e., $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$, so $dxdydz = r^2\sin\theta drd\theta d\phi$ in $0 \leq \theta, \phi \leq \frac{\pi}{2}$ and $0 \leq r \leq a$. Then,

$$\iiint_{V'} x\,dV = \int_0^{\pi/2}\int_0^{\pi/2}\int_0^{a} r\sin\theta\cos\phi\cdot r^2\sin\theta dr d\theta d\phi$$

$$= \int_0^{a} r^3 dr \int_0^{\pi/2} \sin^2\theta\, d\theta \int_0^{\pi/2} \cos\phi\, d\phi = \tfrac{1}{4}a^4 \times \tfrac{1}{4}\pi \times 1 = \tfrac{1}{16}a^4\pi$$

At one thought you may ask, why volume is equal to answer in term of a raise to the power of 4, it seems unreasonable. Refer to the note written at the above, this question is not to find the volume but just a computation of triple integral.